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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

IAM User, Access Keys and AWS Billing Alerts

Course Cloud Infrastructure and Services (25CS2206E)

Skill 1: Creating an IAM User, Generating Access Keys and Configuring Billing Alerts

1. Aim

To create an IAM user with programmatic access only, generate and download the Access Key ID and Secret Access Key, and configure AWS billing alerts (Free Tier usage alerts and a monthly cost budget) so that an email notification is received when usage approaches or exceeds the specified limit.

2. Requirements

Requirement	Details
Web browser	Google Chrome or Microsoft Edge
Website	https://console.aws.amazon.com
AWS account	Active AWS Free Tier account (created in Experiment 1(a))
Login access	Root user, or an account with administrator permissions
Email ID	A valid email address to receive billing and budget alerts
Services used	IAM, Billing and Cost Management, AWS Budgets

3. Theory

AWS Identity and Access Management (IAM) controls who can access an AWS account and what they are allowed to do. Using the root user for everyday work is risky, so a separate IAM user with limited permissions is created instead. AWS also provides tools to track spending so that a free tier account does not produce unexpected charges.

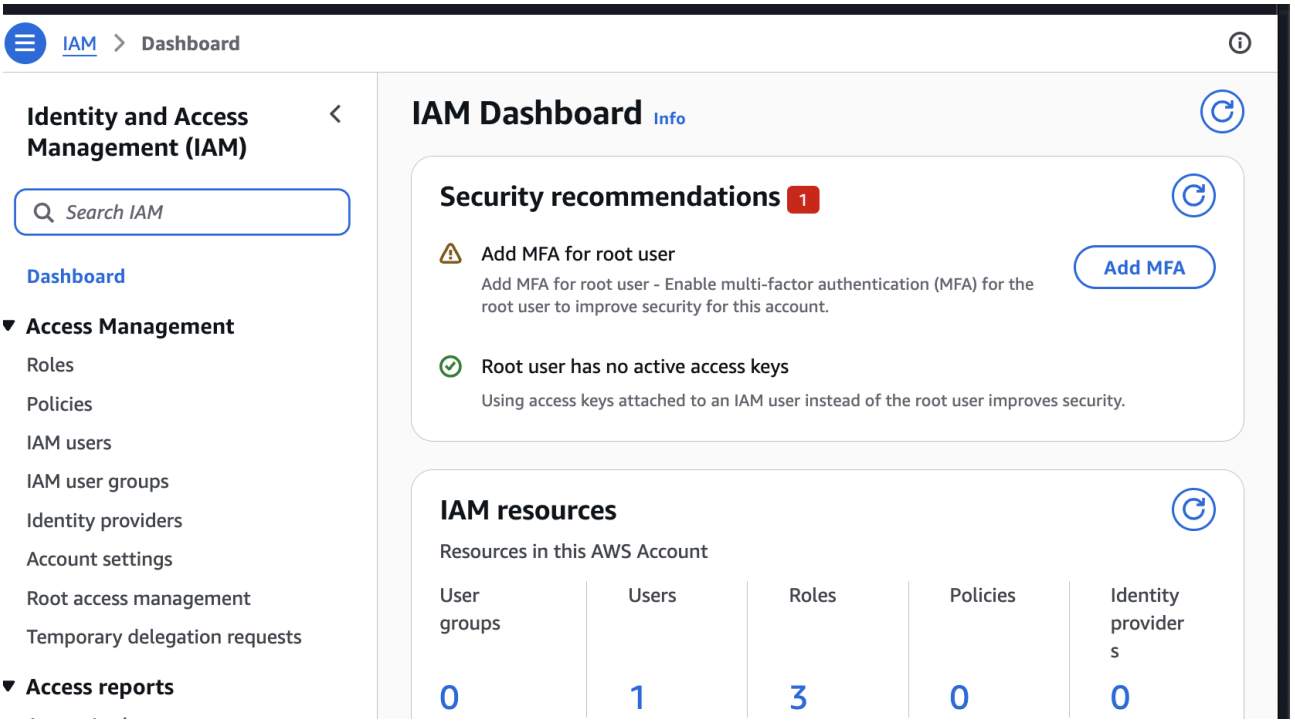
- **IAM user:** An identity created inside the account for a person or an application, with its own permissions.
- **Programmatic access:** Access through the AWS CLI, SDKs or APIs using access keys, with no console sign-in.
- **Access keys:** A pair of an Access Key ID and a Secret Access Key. The secret key is shown only once, so it must be downloaded and stored safely.
- **Permissions policy:** A document that defines which actions the user may perform on which AWS services.
- **Free Tier usage alerts:** Emails sent when usage of a Free Tier service approaches or exceeds its free limit.
- **AWS Budgets:** A tool to set a cost limit and receive alerts when actual spending crosses a chosen threshold.

4. Procedure

Part A: Create an IAM User (Programmatic Access Only)

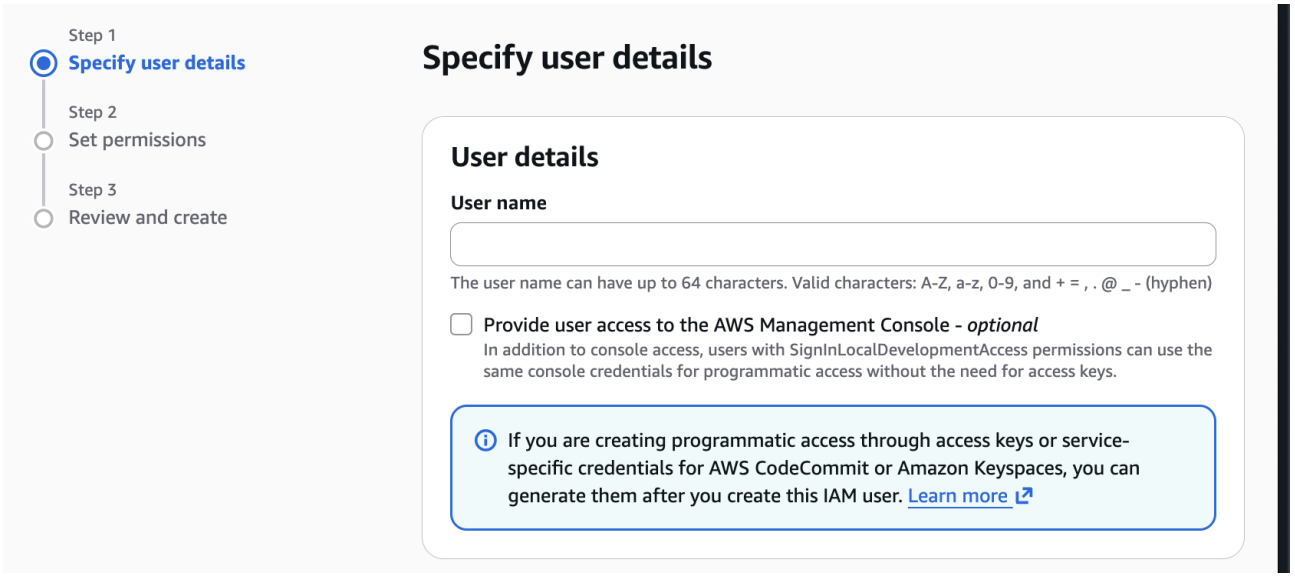
Step 1: Sign in and open IAM

1. Log in to the AWS Management Console using the root user credentials.
2. In the search bar at the top, type **IAM** and select **Identity and Access Management (IAM)**.



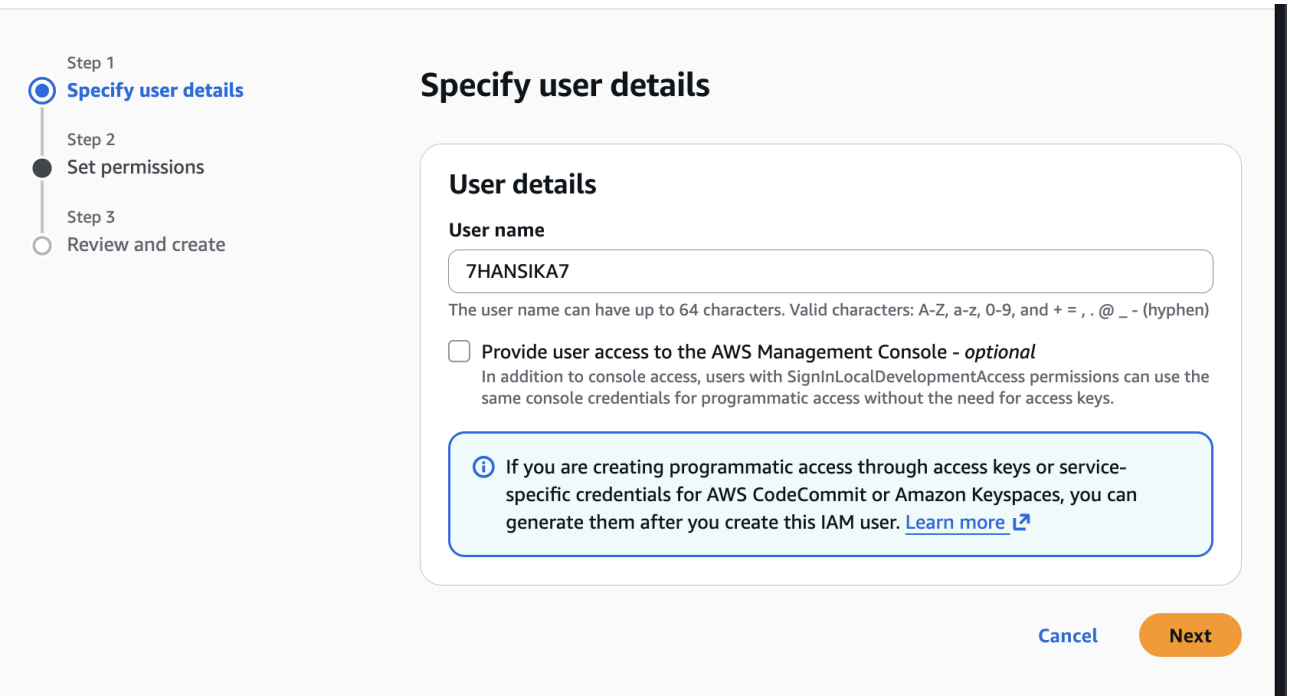
Step 2: Open the Users page

1. From the left navigation panel, select **Users**.
2. Click **Create user**.



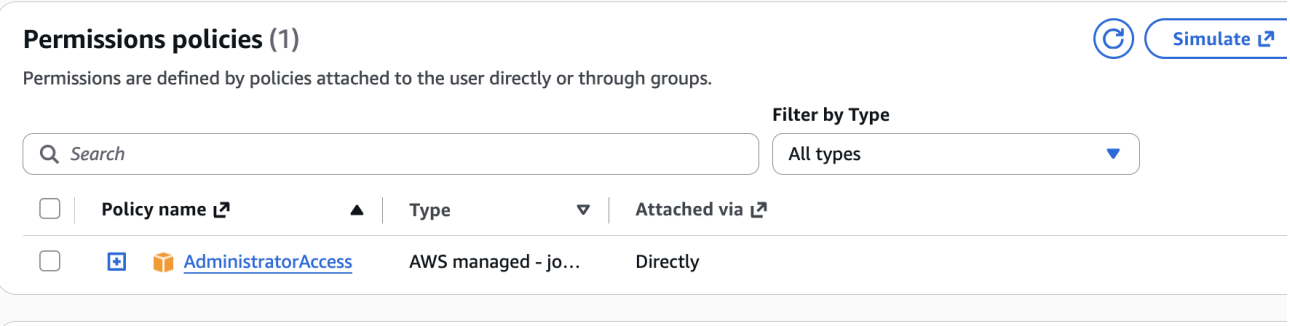
Step 3: Enter the user details

1. Enter a suitable username (e.g., **Student-IAM-User**).
2. Do **not** tick *Provide user access to the AWS Management Console*, because this user needs programmatic access only.
3. Click **Next**.



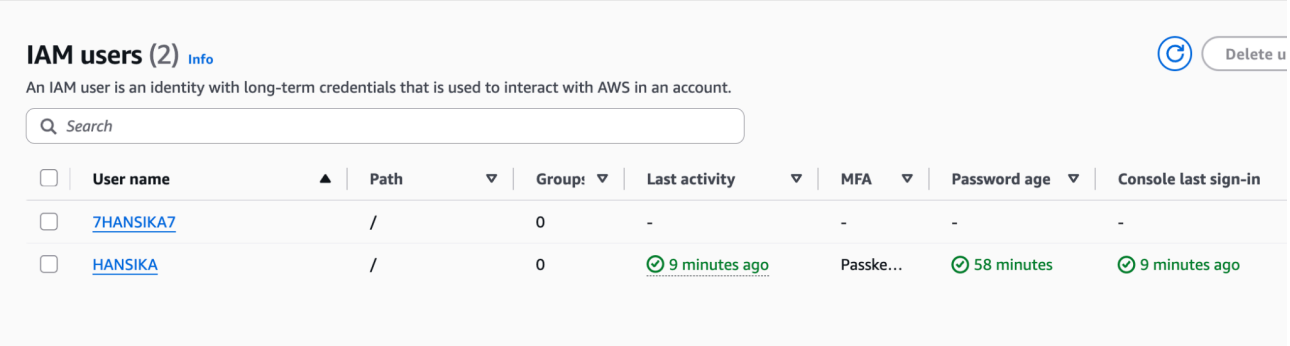
Step 4: Set the permissions

1. Under *Set permissions*, select **Attach policies directly**.
2. Choose the policy required for the tasks to be performed (e.g., **AdministratorAccess**).
3. Click **Next** and review the configuration.



Step 5: Create and verify the user

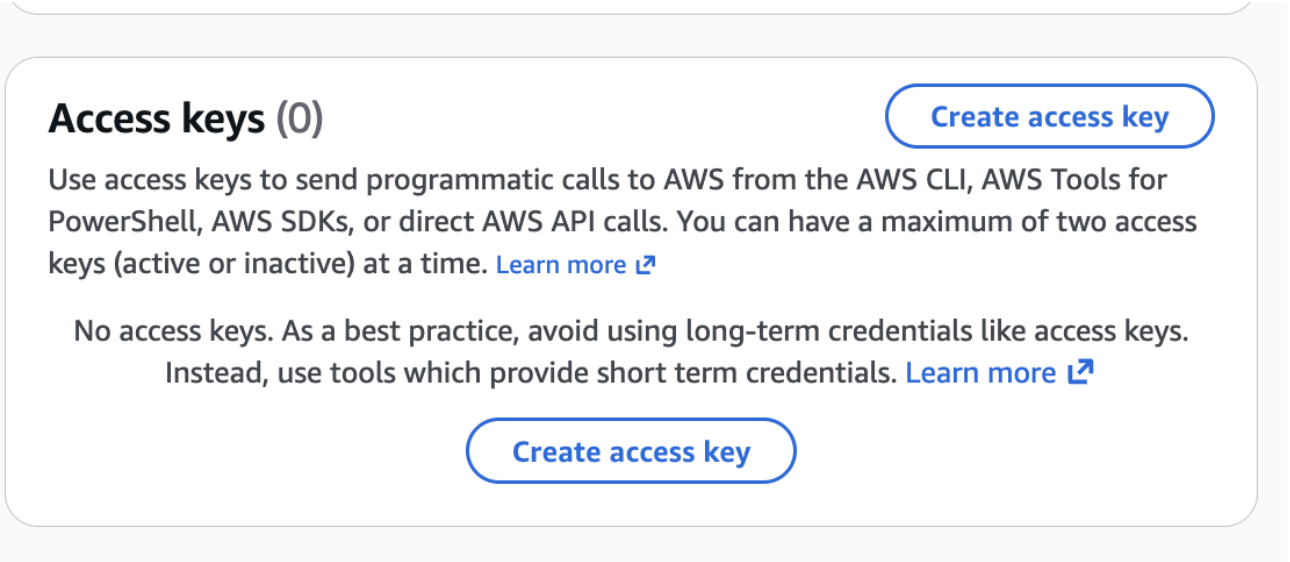
1. Click **Create user**.
2. Check that the new IAM user appears in the **Users** list.



Part B: Generate and Download Access Keys

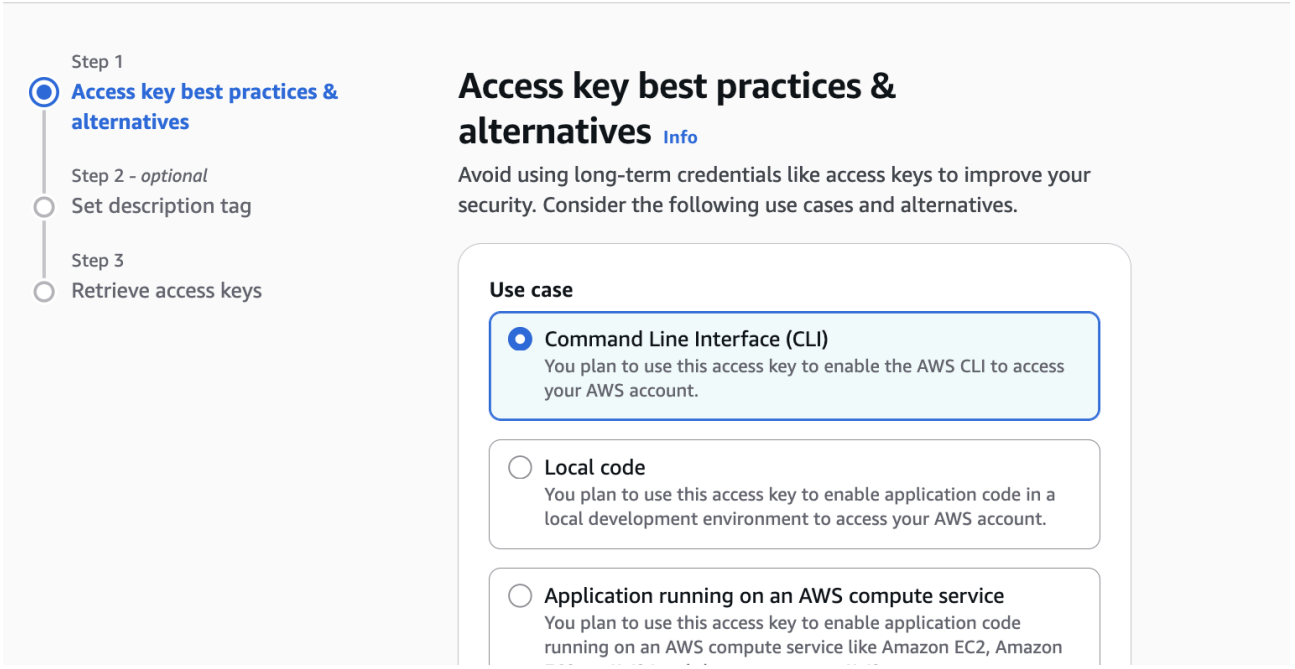
Step 6: Open the Security credentials tab

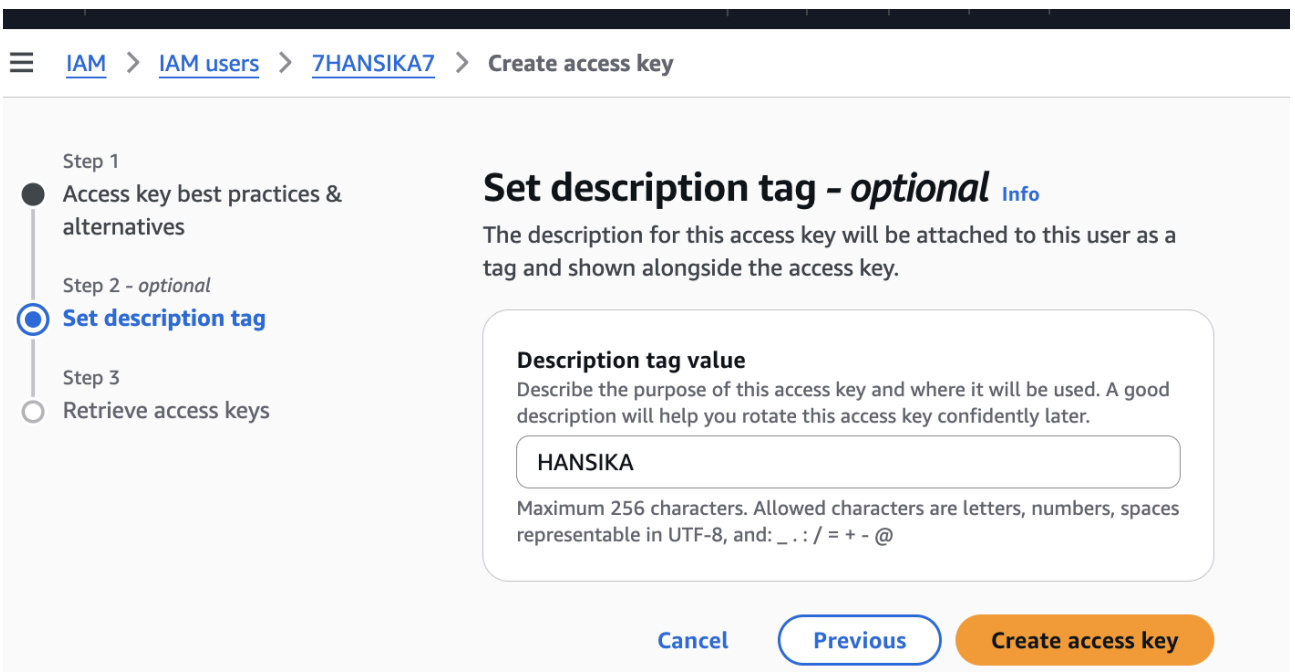
1. In the IAM dashboard, click the newly created IAM user.
2. Open the **Security credentials** tab.
3. Scroll down to the **Access keys** section.



Step 7: Create the access key

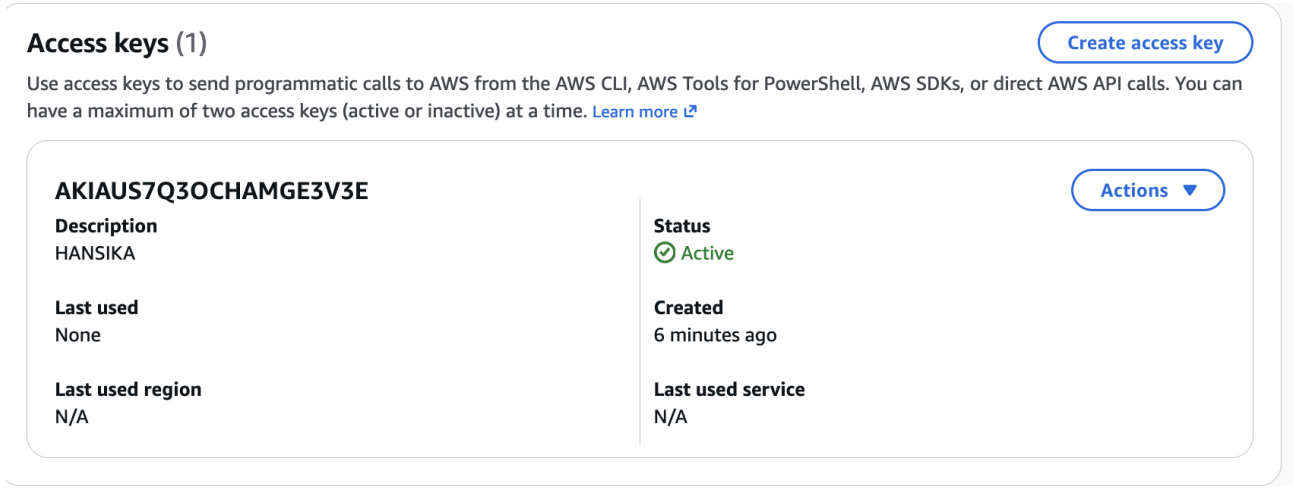
1. Click **Create access key**.
2. Select the use case, such as **Command Line Interface (CLI)**.
3. Tick the confirmation checkbox and click **Next**.
4. Optionally enter a description tag (e.g., **Student-CLI-Access-Key**).
5. Click **Create access key**.





Step 8: Download the credentials

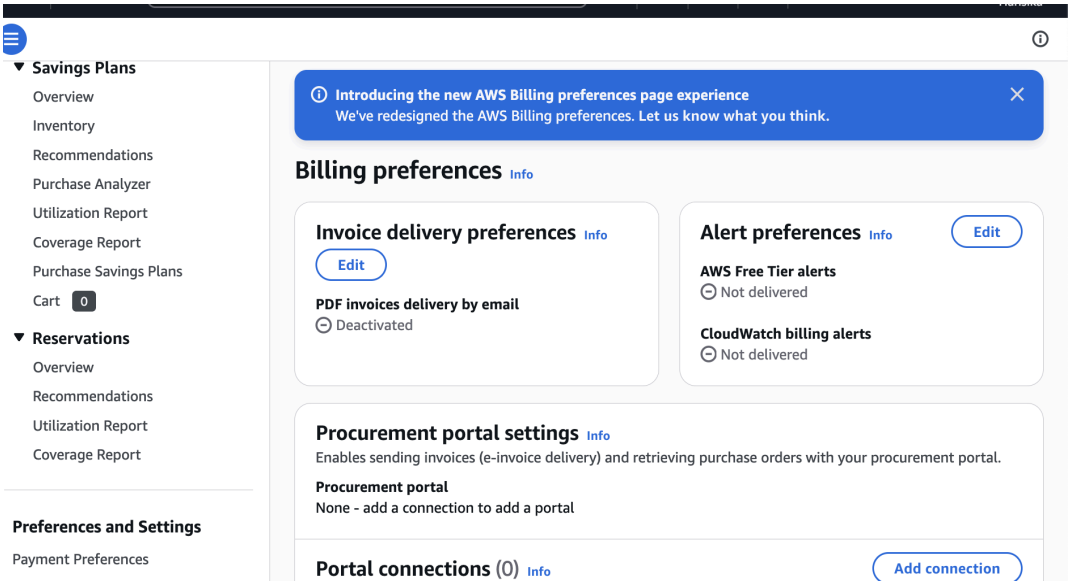
1. The console shows the **Access Key ID** and the **Secret Access Key**.
2. Click **Download .csv file** to save the credentials.
3. Open the CSV file and verify that both keys are present.
4. Click **Done**.



Part C: Configure AWS Free Tier Usage Alerts

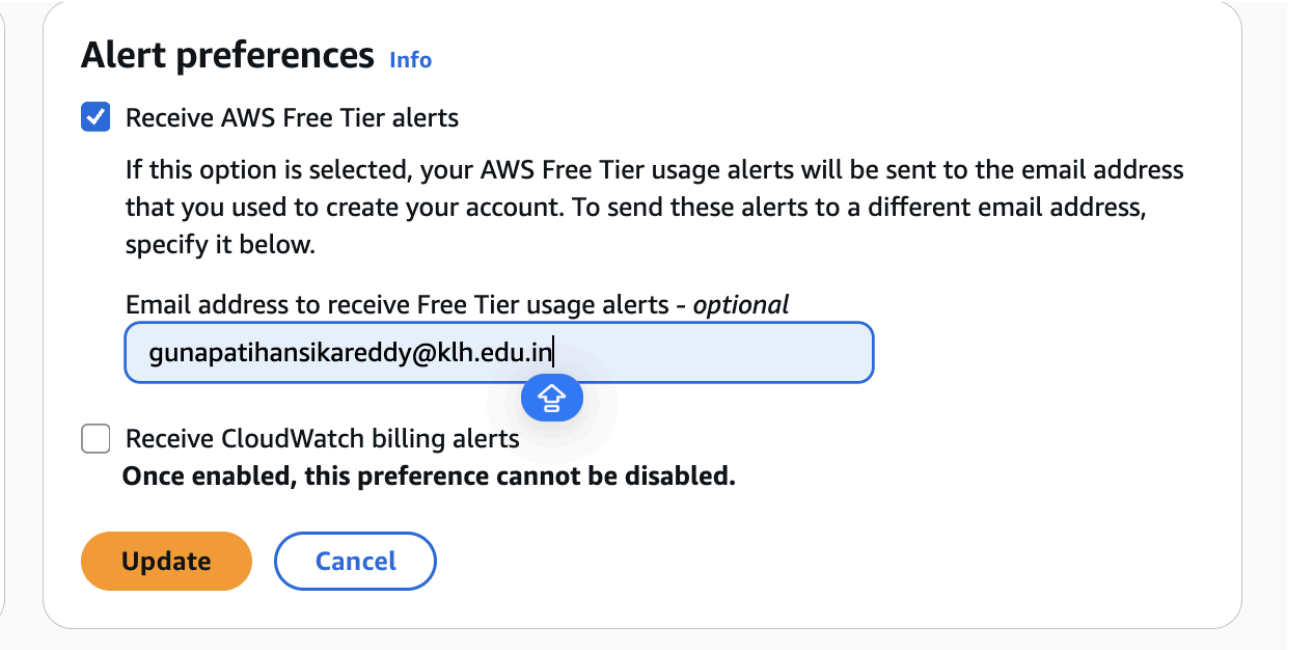
Step 9: Open Billing preferences

1. Log in to the AWS Management Console using the **root user** account.
2. Search for **Billing and Cost Management** and open it.
3. In the left panel, select **Preferences** and then **Billing preferences**.



Step 10: Enable Free Tier usage alerts

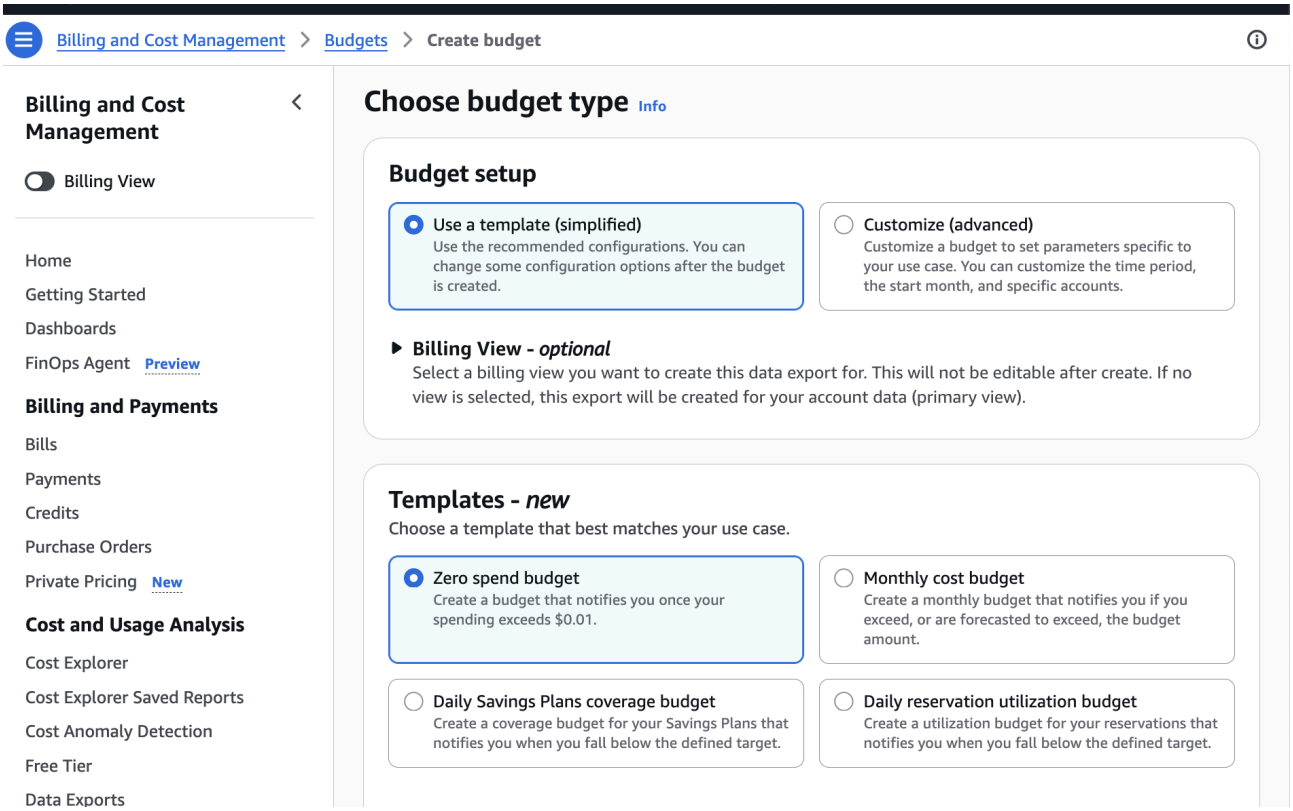
1. Locate the **Alert preferences** section and click **Edit**.
2. Select the checkbox **Receive AWS Free Tier usage alerts**.
3. Under *Additional email address to receive alerts*, enter the email address (if required).
4. Click **Update** and verify that the option is enabled.



Part D: Configure a Billing Alert Using AWS Budgets

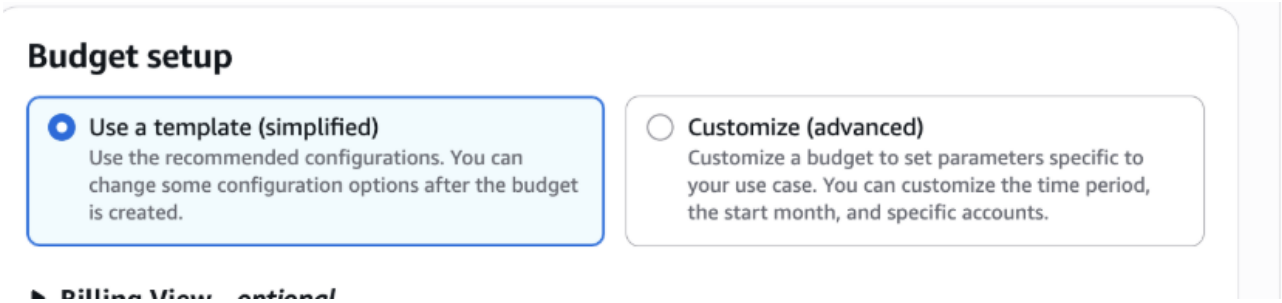
Step 11: Open Budgets and start a new budget

1. In **Billing and Cost Management**, select **Budgets** under *Budgets and Planning*.
2. Click **Create budget**.



Step 12: Choose the budget setup

1. **Option 1 (simple):** Select **Use a template (simplified)** and choose **Zero spend budget**, which notifies you once spending exceeds $0.01. Enter the budget name and email ID, then go to Step 15.
2. **Option 2 (custom):** Select **Customize (advanced)**, choose budget type **Cost budget** and click **Next**.

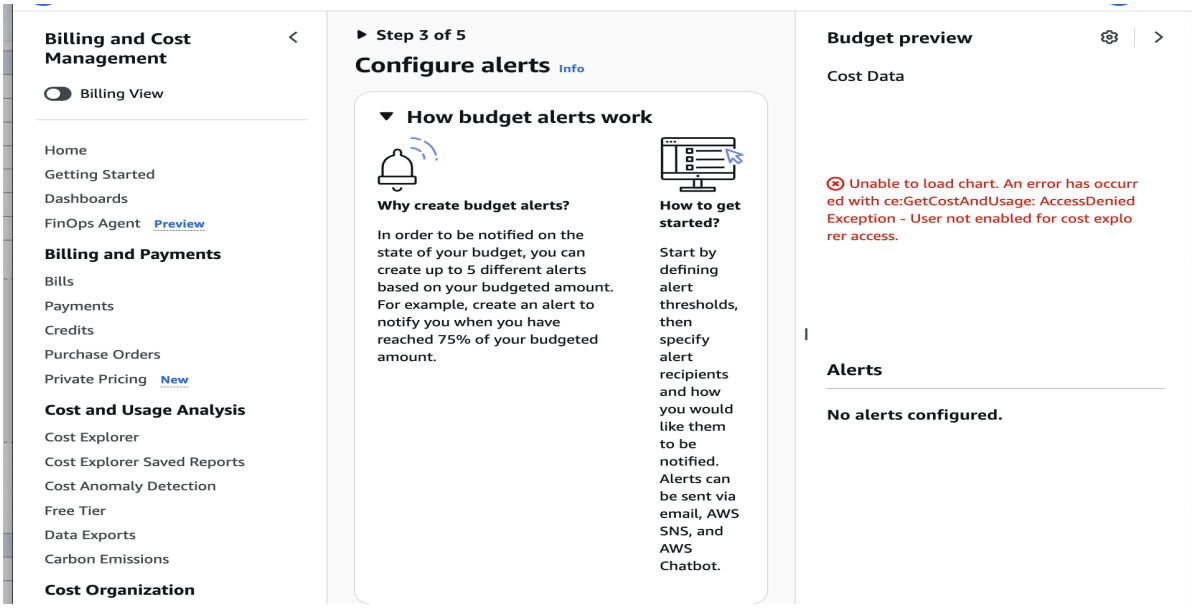


Step 13: Configure the budget details (custom option)

Enter the following details and keep the advanced settings at their default values:

Field	Value
Budget name	AWS-Monthly-Cost-Budget
Period	Monthly
Budget renewal type	Recurring budget
Start month	Current month
Budgeting method	Fixed
Budgeted amount	$0 (example)
Budget scope	All AWS services

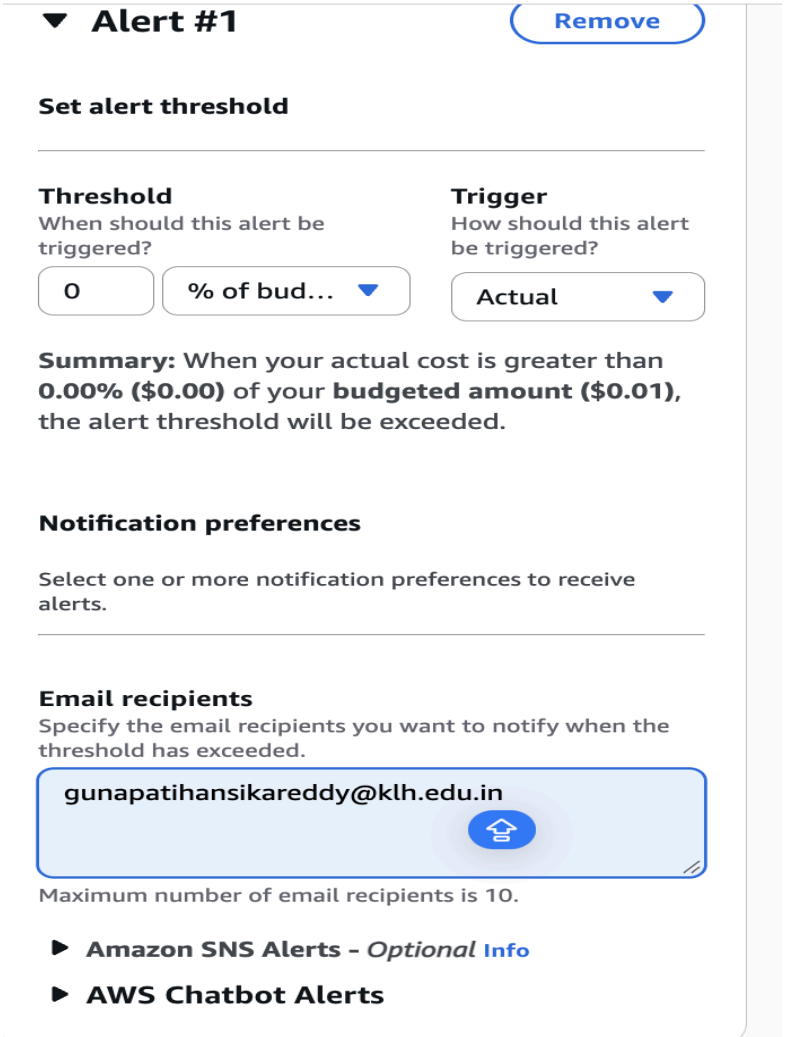
1. Click **Next**.



Step 14: Configure the alert threshold

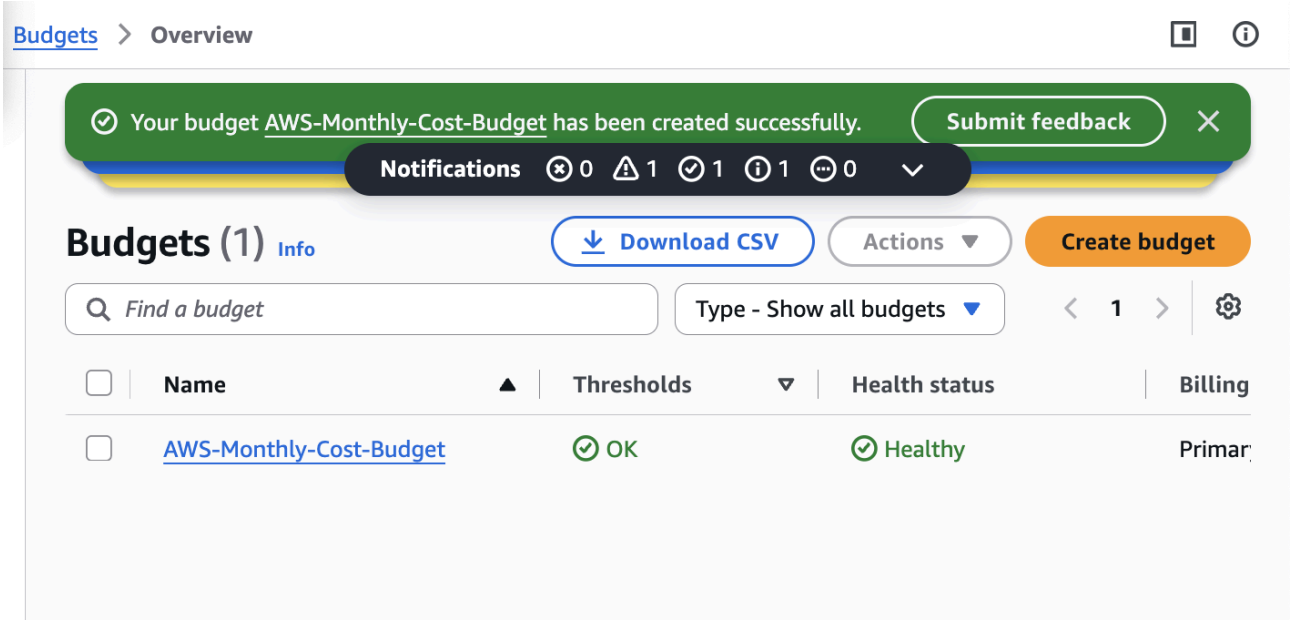
1. Under *Configure alerts*, click **Add an alert threshold**.
2. Enter the alert values given below, then click **Next**.

Field	Value
Threshold	0
Threshold type	% of budgeted amount
Alert trigger	Actual
Email recipients	Your registered email address



Step 15: Skip actions, review and create

1. On the **Attach actions** page, skip the optional actions and click **Next**.
2. Review all the budget details and alert thresholds.
3. Click **Create budget**.



5. Observation

Parameter	Observation
IAM user	Created without console access (programmatic access only)
Permissions	Policy attached directly to the user
Access keys	Access Key ID and Secret Access Key generated and downloaded as a CSV file
Free Tier alerts	Receive AWS Free Tier usage alerts option was enabled
Budget	Monthly cost budget created with an alert at 0% of the budgeted amount (Actual)
Email notification	Alert email goes to the registered email address

6. Result

- An IAM user with programmatic access only was created successfully.
- Access keys were generated and downloaded securely.
- Free Tier usage alerts and a monthly cost budget were configured to send email notifications.

7. Conclusion

An IAM user with its own access keys lets applications and the AWS CLI use the account without exposing the root credentials. Together with Free Tier alerts and AWS Budgets, this gives a safe way to use AWS: access is controlled, the keys are stored securely, and unexpected charges are caught early by email.